

## GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

|                                |                                                              |
|--------------------------------|--------------------------------------------------------------|
| <b>Project</b>                 | FEED for Gas Gathering Pipelines & Associated Infrastructure |
| <b>Client</b>                  | Brass Fertilizer and Petrochemical Company Limited           |
| <b>Originating Company</b>     | Epic Atlantic Limited                                        |
| <b>Document Title</b>          | Pipeline Buoyancy / On Bottom Stability Analysis Report      |
| <b>Document Number</b>         | EA-BFPCL-GPMC-GGPL-FEED- PPL-RPT-003                         |
| <b>Document Revision</b>       | R01                                                          |
| <b>Document Status</b>         | Issued for Review                                            |
| <b>Originator</b>              | E. Omeke                                                     |
| <b>Security Classification</b> | Restricted                                                   |
| <b>Issue Date</b>              | 6-Jun-22                                                     |

## Pipeline Buoyancy / On Bottom Stability Analysis Report

|       |            |                    |            |              |                 |                  |
|-------|------------|--------------------|------------|--------------|-----------------|------------------|
|       |            |                    |            |              |                 |                  |
| R01   | 06.06.22   | Issued for Review  | E. Omeke   | E. Nwachukwu | A.Duru          |                  |
| Rev # | Issue Date | Status Description | Originator | Reviewer     | Approver (Epic) | Approver (BFPCL) |

## Revision History

| Rev No. | Date     | Description of Revision        |
|---------|----------|--------------------------------|
| R01     | 06-06-22 | First Issue for Clients Review |
| R02     |          |                                |
| A01     |          |                                |
|         |          |                                |

| Review Code & Status (to be filled by Client) |                                                                                     |       |
|-----------------------------------------------|-------------------------------------------------------------------------------------|-------|
| <input type="checkbox"/> Code 1               | Approved – Further work may proceed.                                                |       |
| <input type="checkbox"/> Code 2               | Approved with comments. – Incorporate Comments. Up-Rev & Resubmit. Work May Proceed |       |
| <input type="checkbox"/> Code 3               | Reviewed – Incorporate Comments. Up-Rev & Resubmit. Work May Not Proceed            |       |
| <input type="checkbox"/> Code 4               | Retained for information only                                                       |       |
| <input type="checkbox"/> Code 5               | Review Not Required – Do not Resubmit                                               |       |
| Date:                                         | Position:                                                                           | Sign: |

## Revision Philosophy:

- a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
- b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.)  
*for the subsequent project phases, post FEED, it is expected that:*
- c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
- d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
- e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

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## SUMMARY PAGE

|                  |                                                     |                                                        |
|------------------|-----------------------------------------------------|--------------------------------------------------------|
| <b>Client:</b>   | Brass Fertilizer & Petrochemical Company Limited    |                                                        |
| <b>Project:</b>  | Gas Processing & Methanol Complex                   | <b>Project No.</b> 11998A                              |
| <b>Facility:</b> | Gas Gathering Pipelines & Associated Infrastructure | <b>Doc No.</b><br>EA-BFPCL-GPMC-GGPL-FEED- PPL-RPT-003 |

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, MULTIPHASE FLOW, ROW, PTS

## REFERENCES

| Document No.                               | Description                                                       |
|--------------------------------------------|-------------------------------------------------------------------|
| EA-BFPCL-GPMG-GGPL-FEED-PSE-RPT-001        | Upstream Gas Gathering Concept Select & Revalidation Report       |
| EA-BFPCL/BFPP GAS PL/FEED-PRO-RPT-001      | Preliminary Simulation – Based on Desktop Route Survey            |
| EA/SPDC-BFPC UGS-GEP/CSM-RPT-002           | Gas Gathering & Evacuation Pipelines Concept Screening            |
| EA/SPDC-BFPC UGS-GEP/CSM-RPT-001           | Conceptual Study Memorandum                                       |
| EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002        | Basis of Design                                                   |
|                                            | FISAS Draft ESIA Report (November 2020)                           |
| EA-BFPCL-GPMC-GGPL-FEED-PPL-RPT-002        | Wall Thickness Calculation Report                                 |
| EA-BFPCL-GPMC-GGPL-FEED-PPL-ALS-001 to 036 | Pipeline Alignment Sheet                                          |
|                                            | BFPCL Bubouwe - Akabel Topographical Survey Report_Rev.2_08022022 |

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## Abbreviations

| Abbreviation | Meaning                                    |
|--------------|--------------------------------------------|
| API          | American Petroleum Institute               |
| ASME         | American Society of Mechanical Engineers   |
| ASTM         | American Society for Testing and Materials |
| CRA          | Corrosion Resistant Alloy                  |
| CS           | Carbon Steel                               |

# 1. Introduction

## 1.1. Background Information

Brass Fertilizer and Petrochemical Company Limited (BFPCL) is developing a 10,000 metric tonnes per day (MTPD) capacity greenfield methanol production facility in Akabel (Odioma), Brass LGA, Bayelsa state. The plant complex is to be based on natural gas feedstock from 5 SPDC fields in OMLs 33, 35 & 36.

The Project involves the development, construction, and operation of a plant comprising:

- Two trains of 5,000 MTPD methanol plant, producing a total of 10,000 MTPD of methanol
- 340 MMscfd Gas Processing Plant ("GPP"), which will extract NGL from the natural gas feedstock, prior to feeding the lean gas to the methanol plant.
- Gas gathering pipelines and associated manifolds transporting the natural gas from five fields, viz: Bubouwe Bou (BBOU), Elepa (ELEP), Nun River (NUNR), Seibou (SEIB), and Opugbene (OPUG) to the GPP at Akabel.
- Product Export Lines and SBM for the export of the methanol, and NGL of the GPP.
- All offsite, utility, and off-plot facilities (including product storage, internal power generation, steam generation, and export jetty) for the operation of the processing plants.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2.

The focus of the present FEED is on Phase 1 as shown in figure 1 below, while Phase 2 will be implemented later.

TCE is the Project Management Consultant for the project.

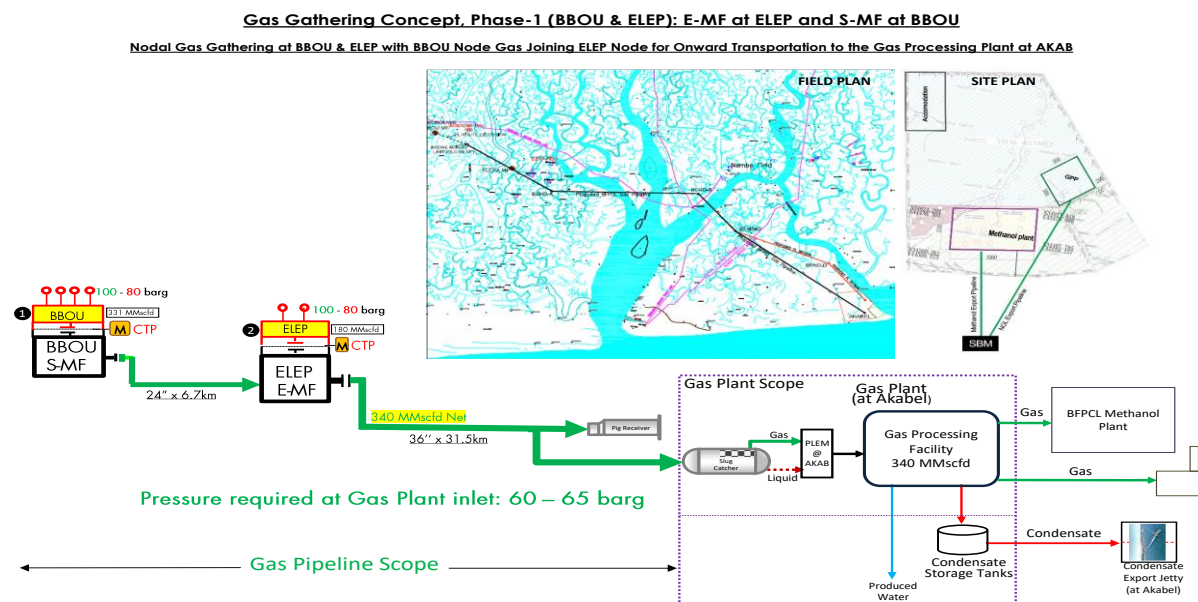


Figure 1: Gas Gathering Concept, Phase 1

## 1.2. Purpose

The purpose of this report is to determine the design requirement for the stability of the 24" API 5L Gr. X70 and 36" API 5L Gr. X70 Pipeline systems of BFPCL Gas gathering pipeline -Phase 1 project by carrying out buoyancy calculation.

The FEED shall fully comply with all specified relevant contractual requirements.

## 2. Regulations, Codes And Standards

| Document Number      | Document Title                                                                               |
|----------------------|----------------------------------------------------------------------------------------------|
| DEP 31.40.00.10-Gen  | Pipeline Engineering (Amendments/Supplement to ISO 13623)                                    |
| DEP. 31.40.20.37-Gen | Line pipe for Critical Service (amendments/supplements to IOGP S-616 AND DNVGL-ST-F101 2019) |
| DEP 37.81.41.30-Gen. | Onshore fabrication of deepwater pipelines, flowlines, risers and their appurtenances        |
| ASME B31.8           | Gas Transmission and Distribution Piping Systems                                             |
| API 5L               | Specification for Line Pipe                                                                  |
| ISO 3183: 2007       | Petroleum and Natural Gas Industries Steel Pipe for pipeline transportation systems.         |
| DNV-RP-F109          | On-bottom Stability Design of Submarine Pipelines                                            |

The latest versions of the regulations, codes, and standards and extant amendments or supplements shall apply. Local laws regulating the oil and gas industry in Nigeria shall be strictly adhered to. The order of precedence for the documents shall be as follows:

- Nigerian Local Standards and Codes
- The Contract
- Project Specifications
- Employer's Standards
- Shell DEPs
- International Codes and Standards

In all cases, the most stringent shall apply except where constraints dictate otherwise.

Conflicts between specifications and referenced Codes and Standards or deviations thereof shall be referred in a Technical Query (TQ) to the EMPLOYER for resolution.

### 3. Line Pipe Data

#### 3.1. 24" Multiphase Pipeline Data

The 24" pipeline design data for the Multiphase Pipeline Project are as shown below

Table 3.1: 24" Multiphase Pipeline Data

| Parameter                               | Description                                |
|-----------------------------------------|--------------------------------------------|
| Design Code                             | ASME B31.8                                 |
| From                                    | BBOU Manifold Station                      |
| To                                      | ELEPA Manifold Station                     |
| Pipeline Length                         | 6.7km                                      |
| Design Life                             | 30 years                                   |
| Process Fluid                           | Condensate / Saturated Hydrocarbon Gas.    |
| Design Pressure                         | 110 barg                                   |
| Operating Pressure                      | 100 barg                                   |
| Max Design Temperature                  | 60°C                                       |
| Min Design Temp (Pipeline)              | 12.7°C                                     |
| Min Design Temperature (Vent Line)      | -13.3°C                                    |
| Line Pipe Material Grade                | API 5L-X70(PLS2) SAWL - 100% radiographed. |
| Installation Temperature                | 23°C                                       |
| Specified Minimum Yield Strength (SMYS) | 483 Mpa                                    |
| NPS                                     | 24"                                        |
| Density                                 | 7,850 kg/m <sup>3</sup>                    |
| Young's Modulus of elasticity           | 2.07 x 10 <sup>5</sup> MPa                 |
| Poisson's ratio                         | 0.3                                        |
| Coefficient of linear thermal expansion | 11.7 x 10 <sup>-6</sup> m/ m/ °C           |
| Ultimate Tensile Strength               | 570 MPa                                    |
| Steel Specific Resistivity              | 0.018 Ω m/mm <sup>2</sup>                  |
| OD / Wall Thickness                     | 610 / 15.88mm                              |
| Design Factor                           | 0.6                                        |
| Corrosion Allowance                     | 3 mm                                       |
| ANSI Class                              | 900#                                       |
| Parent Pipe External Coating            | 3LPE                                       |
| Concrete Coating Thickness              | 60mm                                       |
| Installation Condition                  | Trenching                                  |
| Cathodic Protection                     | Impressed current                          |



### 3.2. 36" Multiphase Pipeline Design Data

The 36" pipeline design data for the Multiphase Pipeline Project are as shown below

Table 3.2: 36" Multiphase Pipeline Data

| PARAMETER                               | DESCRIPTION                                 |
|-----------------------------------------|---------------------------------------------|
| From                                    | ELEPA Manifold Station                      |
| To                                      | AKABEL Station                              |
| Pipeline Length                         | <b>31.5km</b>                               |
| Design Life                             | 30 years                                    |
| Process Fluid                           | Condensate/hydrated Hydrocarbon Gas.        |
| Design Pressure                         | 110 barg                                    |
| Operating Pressure                      | 100 barg                                    |
| Max Design Temperature                  | 60°C                                        |
| Min Design Temp (Pipeline)              | 12.7°C                                      |
| Min Design Temperature                  | -13.3°C                                     |
| Line Pipe Material Grade                | API 5L-X70 (PLS2) SAWL - 100% radiographed. |
| Installation Temperature                | 23°C                                        |
| Specified Minimum Yield Strength        | 483 Mpa                                     |
| NPS                                     | 36"                                         |
| Density                                 | 7,850 kg/m <sup>3</sup>                     |
| Young's Modulus of elasticity           | 2.07 x 10 <sup>5</sup> MPa                  |
| Poisson's ratio                         | 0.3                                         |
| Coefficient of linear thermal expansion | 11.7 x 10 <sup>-6</sup> m/ m/ °C            |
| Ultimate Tensile Strength               | 570MPa                                      |
| Steel Specific Resistivity              | 0.018 Ω m/mm <sup>2</sup>                   |
| OD / Wall Thickness                     | <b>914 / 23.83mm</b>                        |
| Design Factor                           | 0.6                                         |
| Corrosion Allowance                     | 6 mm                                        |
| ANSI Class                              | 900#                                        |
| Parent Pipe External Coating            | 3LPE                                        |
| Concrete Coating Thickness              | <b>85mm</b>                                 |
| Installation Condition                  | Trenching                                   |
| Cathodic Protection                     | Impressed current                           |

## 4. Pipeline Characteristics

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### 4.1. Material Characteristics

The Gas Pipeline material is carbon steel manufactured to API 5L grade X70. The mechanical properties are as listed in the tables in section 3 above.

### 4.2. Pipeline Coating Characteristics

The pipeline is externally coated with 3-Layer Polyethylene (3-LPE) coating and concrete weight coating to provide stability and mechanical protection. The following properties will be used in the calculations:

$r_{cor}$  - 3-LPE coating density: 942 kg/m<sup>3</sup> (9.24 kN/m<sup>3</sup>)

$t_{cor}$  - 3-LPE coating thickness 3.0 mm

$r_{cc}$  - Concrete coating density 3040 kg/m<sup>3</sup> (29.81 kN/m<sup>3</sup>)

The minimum thickness of the concrete coating shall not be less than **40 mm**.

## 5. Site and Environmental Conditions (Data from ESIA Report)

|                          |                                                                                                                                                                                                                                      |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Air Temperature          | Average daytime 32oC; rainy season 28oC; night time 23oC                                                                                                                                                                             |
| Weather                  | Dry Season (Harmattan); November – January; the atmosphere is laden with extremely fine dust (5 microns particle size)<br>Wet Season: heavy rainfall, two peaks, August- 245mm, September-331 mm                                     |
| Relative Humidity        | Mean 81%; Minimum 69% & Maximum 88%                                                                                                                                                                                                  |
| Wind Speed and Direction | monthly mean 8.5km/h; varies from 6.2km/h to 9.6km/h<br>The south-westerly wind is 60% predominant during the wet season (March - October).<br>The north-easterly wind is 32% predominant during the dry season (November– February) |
| Seismicity               | The project area is classified as Seismic Zone 0                                                                                                                                                                                     |

## 6. Soil Geotechnical Data

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There are no geotechnical soil parameters shown on the survey report issued by Brone Positioning and Survey Ltd with respect to this Project. The Geotechnical data presented here below are extracted from the Survey Report of a similar project carried out within the swampy terrain of the Niger Deltar region of Nigeria. These data should be reconfirmed in detailed engineering when a more robust geotechnical survey is carried out.

The following parameters will be used in the calculations:

|           |                                                                                      |
|-----------|--------------------------------------------------------------------------------------|
| $r_d$     | Dry soil density : 785 kg/m <sup>3</sup> (7.7 kN/m <sup>3</sup> )                    |
| $r_{sat}$ | Saturated soil density : 1468 kg/m <sup>3</sup> (14.4 kN/m <sup>3</sup> )            |
| $r_w$     | Water density: 1150 kg/m <sup>3</sup> (11.28 kN/m <sup>3</sup> )                     |
| $r_{sub}$ | $(r_{sat}-r_w)$ - Submerged density: 318 kg/m <sup>3</sup> (3.12 kN/m <sup>3</sup> ) |

### 6.1. Ground Water

Ground water level fluctuates due to seasonal, tidal, and other factor. In the swampy areas, the water table completely covers the ground, especially in the rainy season, therefore for the calculation purpose the soil has been considered completely immersed in water.

### 6.2. Water Current

Since the major river crossings will be crossed by using Horizontal Directional Drilling technologies, the water current velocities have not been considered in the stability calculations.

## 7. Pipeline Buoyancy Analysis

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Since the pipeline is completely buried in natural soil all along the pipeline route, elaborate on-bottom stability analysis is not considered. Basically, three analyses will be carried out;

- during installation phase with trench open,
- Pipe installed with trench backfilled and
- during operation of the pipeline.

The possibility of pipe flotation exists when the pipeline is constructed in areas which will be inundated, such as watercourse crossings and swampy areas. The buoyancy force is the upward force acting on a body when submerged in a fluid. The body may float on the fluid if the weight of the body is smaller than the upward force.

More generally, floatation may occur if the buoyancy force is larger than the sum of all downward forces acting on the pipeline. In principle, to prevent floatation, the pipeline must be heavier than the upward force.

When such conditions exist, all necessary measures to ensure that when the pipe is laid at the correct depth, in accordance with construction drawings, it rests at that depth and does not move vertically in response to negative buoyancy should be evaluated.

The buoyancy of a pipeline depends upon the weight of the pipe, the weight of the volume of water displaced by the pipe, the weight of the fluid load carried by the pipe and the weight of the backfill.

The cases listed below were evaluated to assure pipeline stability:

- i. Installation phase [CASE I]
  - pipeline is empty.
  - wall thickness includes corrosion allowance.
  - the trench is open and hence no backfill is present.
- ii. Buried [CASE II]
  - pipeline is empty.
  - wall thickness includes corrosion allowance.
  - the trench is backfilled.
- iii. Operation phase [CASE III]
  - fluid is flowing inside pipeline.
  - corrosion allowance is not considered in the wall thickness.

The buoyancy factor is defined as the ratio between the mass of a known length of pipe with sealed ends, and the mass of the volume of water that is displaced

when the pipe is fully submerged.

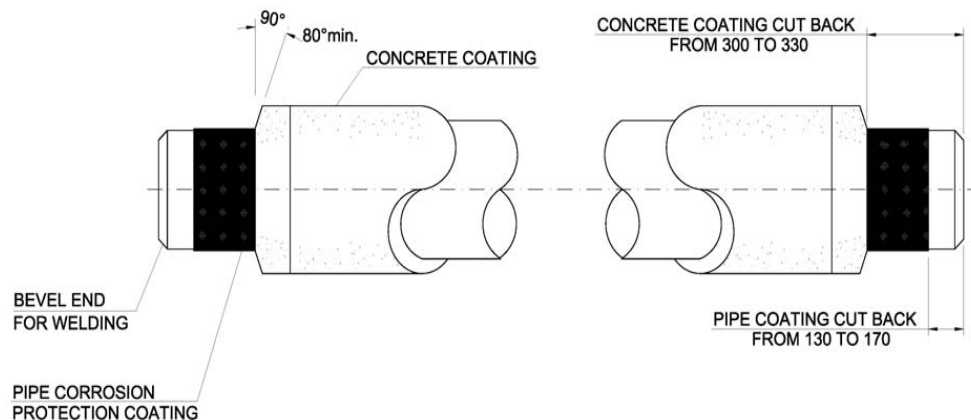
If the upward force is smaller than the weight, there will be no risk of floatation and the ratio is greater than 1. In this condition the pipe will have negatively buoyant.

The safety factor  $F$ , is given by the following ratio:

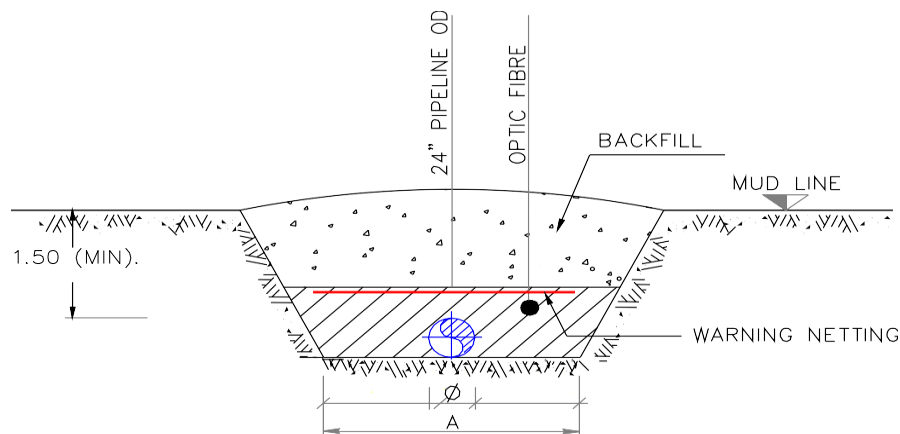
$$F = \frac{\text{mass of body} \times (\text{Pipe} + \text{Anticorrosion Coating} + \text{Concrete Coating})}{\text{volume of body} \times \text{density of slurry}}$$

To ensure the bottom stability of the pipeline under all conditions, the safety factor  $F$  should be greater than 1.25. To achieve a safety factor greater than 1.25, a continuous concrete coating shall be applied to the outer diameter of the pipe, over the external anti-corrosion coating.

The thickness of the concrete coating is dependent upon the pipe wall thickness, the density of the displaced fluid (short and long term), and the density of the concrete coating.



Typical sample of Concrete coated Pipe

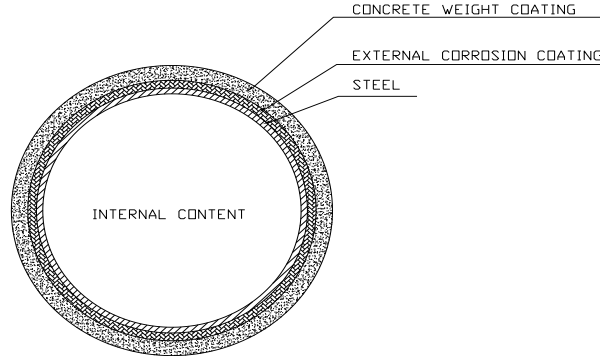


Typical section of a buried pipeline in a trench

Below presents the definitions of the acronyms applied in the calculations and used in the spread sheets while attached 1 and Attachment 2 below are the spread sheet calculations done for the

verification of downward forces (the weight of pipeline) and upward forces (buoyancy) acting on the pipeline which are used in the determination of the required thickness of concrete weight coating.

### PIPE BUOYANCY CALCULATION



**Pipeline Cross Section**

The weight of the components (in air) are calculated as follows:

- |                                                                   |                                                  |
|-------------------------------------------------------------------|--------------------------------------------------|
| 1. Bare Pipe Weight ( $W_1$ ) (Installation Stage)                | $W_1 = \pi (D - t) t \rho_s$                     |
| 2. Bare Pipe Weight ( $W_{1L}$ ) (After Long Period of Operation) | $W_{1L} = \pi (D - (t - CA)) (t - CA) \rho_s$    |
| 3. Corrosion Coat Weight ( $W_2$ )                                | $W_2 = \pi (D + t_c) t_c \rho_c$                 |
| 4. Concrete Coat Weight ( $W_3$ )                                 | $W_3 = \pi (D + 2t_c + t_{cc}) t_{cc} \rho_{cc}$ |
| 5. Weight of Product ( $W_4$ )                                    | $W_4 = \pi/4 (D - 2t)^2 \rho_{pr}$               |
| 6. Backfill Weight ( $W_5$ )                                      | $W_5 = \{(d_p)(\rho_{sub})(D_T)\}$               |
| 7. Weight of Displaced Water ( $W_6$ )                            | $W_6 = \pi/4 (D_T)^2 \rho_{sw}$                  |

Three cases are of interest:

- Installation – Consideration 1, pipeline empty, no backfilling and no loss of corroded material.  
Consideration 2, pipeline empty, with backfilling and no loss of corroded material.
- Operational – Consideration 1, pipeline with content, no backfilling and no metal loss to corrosion included.  
Consideration 2, pipeline with content, with backfilling and no metal loss to corrosion included.
- Operational End of Life – Consideration 1, pipeline empty, with backfilling and corrosion allowance has been lost to corrosion.  
Consideration 2, pipeline with content, with backfilling and corrosion allowance has been lost to corrosion.

Pipeline buoyancy,

$$W_6 = \pi/4 (D_T)^2 \rho_w$$

#### Nomenclature

1.  $D$  :- Outside Diameter
2.  $D_T$  :- Total Outside Diameter including corrosion and concrete coating
3.  $t$  :- Pipe Wall Thickness (Installation Stage)
4.  $t_c$  :- Corrosion Coating Thickness
5.  $t_{cc}$  :- Concrete Coating Thickness
6.  $CA$  :- Corrosion Allowance
7.  $\rho_s$  :- Density of Steel
8.  $\rho_c$  :- Corrosion Coating Density
9.  $\rho_{cc}$  :- Concrete Coating Density
10.  $\rho_{pr}$  :- Concrete Coating Density
11.  $\rho_{sw}$  :- Water Specific Density
12.  $\rho_w$  :- Density of Water
13.  $\rho_{sub}$  :- Submerged Density
14.  $b_p$  :- Burial depth



## 8. Conclusion

The Summary of the analysis for the two pipeline systems is presented in the table below

| STABILITY ANALYSIS SUMMARY |                       |                          |                          |                 |              |              |              |                     |                                 |
|----------------------------|-----------------------|--------------------------|--------------------------|-----------------|--------------|--------------|--------------|---------------------|---------------------------------|
| Material                   | Nominal Diameter (in) | Pipe Wall Thickness (mm) | Corrosion Allowance (mm) | CASE 1 (No CWC) | CASE 1 (CWC) | CASE 2 (CWC) | CASE 3 (CWC) | Over weight Measure | Concrete Coating Thickness (mm) |
|                            |                       |                          |                          | Safety factor   |              |              |              |                     |                                 |
| CS API 5L X70              | 24"                   | 15.88                    | 3.00                     | 0.69            | 1.28         | 1.55         | 1.49         | YES                 | 60.00                           |
|                            | 36"                   | 23.83                    | 6.00                     | 0.69            | 1.26         | 1.74         | 1.66         | YES                 | 85.00                           |

In the Spread sheets, data are clearly divided in two columns, for the INPUT and the OUTPUT values. The input data are:

- the pipe sizes (outside diameter, wall thickness);
- the water density;
- the density of the fluid transported;
- the density of the soil in the trench (both dry and saturated);
- the height of water over the pipeline.
- the pipeline cover depth.

The output column contains the results of the calculations; with details of conditions, acting forces and the required thickness of concrete needed for each meter of pipeline.

### 8.1. Summary and Recommendation

Based on the analysis carried out (see appendix 1 and 2) the Factor of Safety for Buoyancy Calculation are summarized in the table above

The analysis shows that the 24" gas pipeline require a 60mm thick Concrete weight coating system to increase the self-weight to meet a Factor of safety of 1.28 during installation phase only. This implies that if the construction of the pipeline would be undertaken during the dry season when the water table could be reasonably low or below the bottom of the trench, the backfill weight is sufficient to assure a safety factor greater than the minimum required during the installation stage hence the pipeline can be installed without the need for Concrete weight coating.

This recommendation shall be carefully evaluated during the detailed engineering phase as the pipeline route survey result shows that there are several river and creek crossings,

seasonal and permanent swampy areas, which could make it impossible to achieve a water free trenching even in the dry season. Other anti-buoyancy measures such as the removal of water from the trenches using pumps should also be evaluated.

The 36" Pipeline system did not achieve stability in all the three cases of installation, buried and operational phases without additional weight. It requires a concrete weight coating thickness of 85mm for a minimum safety factor of 1.26 to ensure stability.

## 9. Attachment 1: Pipe Buoyancy Calculation For 24", API- 5L X70, 15.88 mm Thick

---

| 24", API- 5L X70, 15.88 mm THK PIPELINES (WITHOUT CONCRETE COATING)            |                                                                   |   |        |                   |
|--------------------------------------------------------------------------------|-------------------------------------------------------------------|---|--------|-------------------|
| PIPE BUOYANCY CALCULATION                                                      |                                                                   |   |        |                   |
| <b>INPUT DATA</b>                                                              |                                                                   |   |        |                   |
| Outside Diameter                                                               | D                                                                 | = | 610    | mm                |
| Pipe Wall Thickness (Installation Stage)                                       | t                                                                 | = | 15.88  | mm                |
| Pipe Wall Thickness (After Long Period of Operation)                           | t <sub>2</sub>                                                    | = | 12.88  | mm                |
| Corrosion Coating Thickness                                                    | t <sub>c</sub>                                                    | = | 3      | mm                |
| Concrete Coating Thickness                                                     | t <sub>cc</sub>                                                   | = | 0      | mm                |
| Corrosion Allowance (CA)                                                       |                                                                   | = | 3      | mm                |
| Total Diameter                                                                 | D <sub>totp</sub>                                                 | = | 616    | mm                |
| Density of Steel                                                               | ρ <sub>s</sub>                                                    | = | 7850   | kg/m <sup>3</sup> |
| Corrosion Coating Density                                                      | ρ <sub>c</sub>                                                    | = | 942    | kg/m <sup>3</sup> |
| Concrete Coating Density                                                       | ρ <sub>cc</sub>                                                   | = | 0      | kg/m <sup>3</sup> |
| Concrete Coating Cut-back                                                      | L                                                                 | = | 0.33   | m                 |
| Product Density                                                                | ρ <sub>pr</sub>                                                   | = | 80     | kg/m <sup>3</sup> |
| Density of water                                                               | ρ <sub>w</sub>                                                    | = | 1150   | kg/m <sup>3</sup> |
| Factor of safety                                                               | F <sub>s</sub>                                                    | = | 1.25   |                   |
| Burial depth                                                                   | d <sub>b</sub>                                                    | = | 1.5    | m                 |
| <b>SOIL DATA</b>                                                               |                                                                   |   |        |                   |
| Water Specific Density                                                         |                                                                   | = | 11.28  | kN/m <sup>3</sup> |
| Dry Soil Density                                                               | ρ <sub>d</sub>                                                    | = | 7.7    | kN/m <sup>3</sup> |
| Saturated Soil Density                                                         | ρ <sub>sat</sub>                                                  | = | 14.40  | kN/m <sup>3</sup> |
| Submerged Density                                                              | ρ <sub>sub</sub>                                                  | = | 3.12   | kN/m <sup>3</sup> |
| <b>OUTPUT</b>                                                                  |                                                                   |   |        |                   |
| Bare Pipe Weight (W <sub>1</sub> ) (Installation Stage)                        |                                                                   | = | 2.283  | KN/m              |
| Bare Pipe Weight (W <sub>1L</sub> ) (After Long Period of Operation)           |                                                                   | = | 1.861  | KN/m              |
| Corrosion Coat Weight (W <sub>2</sub> )                                        |                                                                   | = | 0.053  | KN/m              |
| Concrete Coat Weight (W <sub>3</sub> )                                         |                                                                   | = | 0.000  | KN/m              |
| Weight of Product (W <sub>4</sub> )                                            |                                                                   | = | 0.206  | KN/m              |
| Backfill Weight (W <sub>5</sub> )                                              |                                                                   | = | 2.881  | KN/m              |
| Weight of Displaced Water (W <sub>6</sub> )                                    |                                                                   | = | 3.362  | KN/m              |
| Buoyancy of pipe (W <sub>b</sub> )                                             |                                                                   | = | 3.362  | KN/m              |
| Total weight of pipe (W <sub>p</sub> ) (Installation Stage)                    | W <sub>1</sub> +W <sub>2</sub> +W <sub>3</sub>                    | = | 2.336  | KN/m              |
| Total weight of pipe (W <sub>pl</sub> ) (After Long Period of Operation)       | W <sub>1L</sub> +W <sub>2</sub> +W <sub>3</sub>                   | = | 1.914  | KN/m              |
| Submerged weight of pipe (W <sub>sb</sub> )                                    | W <sub>p</sub> - W <sub>b</sub>                                   | = | -1.026 | KN/m              |
| <b>FACTOR OF SAFETY</b>                                                        |                                                                   |   |        |                   |
| <b>[ CASE I ]</b>                                                              |                                                                   |   |        |                   |
| Empty Pipe without backfilling (Installation Stage)                            | W <sub>p</sub> /W <sub>b</sub>                                    | = | 0.69   | 'FAIL'            |
| Pipe with product without backfilling                                          | (W <sub>p</sub> +W <sub>4</sub> )/W <sub>b</sub>                  | = | 0.76   | 'FAIL'            |
| <b>[ CASE II ]</b>                                                             |                                                                   |   |        |                   |
| Empty Pipe with backfilling (Installation Stage)                               | (W <sub>p</sub> +W <sub>5</sub> )/W <sub>b</sub>                  | = | 1.55   | PASS              |
| Empty Pipe with backfilling (After Long Period of Operation)                   | (W <sub>pl</sub> +W <sub>5</sub> )/W <sub>b</sub>                 | = | 1.43   | PASS              |
| <b>[ CASE III ]</b>                                                            |                                                                   |   |        |                   |
| Pipe with product and backfilling                                              | (W <sub>pl</sub> +W <sub>5</sub> +W <sub>4</sub> )/W <sub>b</sub> | = | 1.49   | PASS              |
| <b>UPWARD FORCE</b>                                                            |                                                                   |   |        |                   |
| Empty Pipe                                                                     | W <sub>b</sub> -W <sub>p</sub>                                    | = | 1.03   | KN/m              |
| Pipe with Product                                                              | W <sub>b</sub> -(W <sub>4</sub> +W <sub>p</sub> )                 | = | 0.82   | KN/m              |
| <b>RESULT: THE PIPELINE DESIGN IS INADEQUATE, CONCRETE COATING IS REQUIRED</b> |                                                                   |   |        |                   |

| 24", API- 5L X70, 15.88mm THK PIPELINES (WITH CONCRETE COATING)          |                                                                   |   |       |                   |
|--------------------------------------------------------------------------|-------------------------------------------------------------------|---|-------|-------------------|
| PIPE BUOYANCY CALCULATION                                                |                                                                   |   |       |                   |
| <b>INPUT DATA</b>                                                        |                                                                   |   |       |                   |
| Outside Diameter                                                         | D                                                                 | = | 610   | mm                |
| Pipe Wall Thickness (Installation Stage)                                 | t                                                                 | = | 15.88 | mm                |
| Pipe Wall Thickness (After Long Period of Operation)                     | t <sub>2</sub>                                                    | = | 12.88 | mm                |
| Corrosion Coating Thickness                                              | t <sub>c</sub>                                                    | = | 3     | mm                |
| Concrete Coating Thickness                                               | t <sub>cc</sub>                                                   | = | 60    | mm                |
| Corrosion Allowance (CA)                                                 | (CA)                                                              | = | 3     | mm                |
| Total Diameter                                                           | D <sub>totp</sub>                                                 | = | 736.0 | mm                |
| Density of Steel                                                         | ρ <sub>s</sub>                                                    | = | 7850  | kg/m <sup>3</sup> |
| Corrosion Coating Density                                                | ρ <sub>c</sub>                                                    | = | 942   | kg/m <sup>3</sup> |
| Concrete Coating Density                                                 | ρ <sub>cc</sub>                                                   | = | 3040  | kg/m <sup>3</sup> |
| Concrete Coating Cut-back                                                | L                                                                 | = | 0.33  | m                 |
| Product Density                                                          | ρ <sub>pr</sub>                                                   | = | 80    | kg/m <sup>3</sup> |
| Density of water                                                         | ρ <sub>w</sub>                                                    | = | 1150  | kg/m <sup>3</sup> |
| Factor of safety                                                         | F <sub>s</sub>                                                    | = | 1.25  |                   |
| Burial depth                                                             | d <sub>b</sub>                                                    | = | 1.50  | m                 |
| <b>SOIL DATA</b>                                                         |                                                                   |   |       |                   |
| Water Specific Density                                                   |                                                                   | = | 11.28 | kN/m <sup>3</sup> |
| Dry Soil Density                                                         | ρ <sub>d</sub>                                                    | = | 7.7   | kN/m <sup>3</sup> |
| Saturated Soil Density                                                   | ρ <sub>sat</sub>                                                  | = | 14.40 | kN/m <sup>3</sup> |
| Submerged Density                                                        | ρ <sub>sub</sub>                                                  | = | 3.12  | kN/m <sup>3</sup> |
| <b>OUTPUT</b>                                                            |                                                                   |   |       |                   |
| Bare Pipe Weight (W <sub>1</sub> ) (Installation Stage)                  |                                                                   | = | 2.283 | KN/m              |
| Bare Pipe Weight (W <sub>1L</sub> ) (After Long Period of Operation)     |                                                                   | = | 1.861 | KN/m              |
| Corrosion Coat Weight (W <sub>2</sub> )                                  |                                                                   | = | 0.053 | KN/m              |
| Concrete Coat Weight (W <sub>3</sub> )                                   |                                                                   | = | 3.800 | KN/m              |
| Weight of Product (W <sub>4</sub> )                                      |                                                                   | = | 0.206 | KN/m              |
| Backfill Weight (W <sub>5</sub> )                                        |                                                                   | = | 3.443 | KN/m              |
| Weight of Displaced Water (W <sub>6</sub> )                              |                                                                   | = | 4.800 | KN/m              |
| Buoyancy of pipe (W <sub>b</sub> )                                       |                                                                   | = | 4.800 | KN/m              |
| Total weight of pipe (W <sub>p</sub> ) (Installation Stage)              | W <sub>1</sub> +W <sub>2</sub> +W <sub>3</sub>                    | = | 6.136 | KN/m              |
| Total weight of pipe (W <sub>pl</sub> ) (After Long Period of Operation) | W <sub>1L</sub> +W <sub>2</sub> +W <sub>3</sub>                   | = | 5.714 | KN/m              |
| Submerged weight of pipe (W <sub>sb</sub> )                              | W <sub>p</sub> - W <sub>b</sub>                                   | = | 1.336 | KN/m              |
| <b>FACTOR OF SAFETY</b>                                                  |                                                                   |   |       |                   |
| <b>[ CASE I ]</b>                                                        |                                                                   |   |       |                   |
| Empty Pipe without backfilling (Installation Stage)                      | W <sub>p</sub> /W <sub>b</sub>                                    | = | 1.28  | PASS              |
| Pipe with product without backfilling                                    | (W <sub>p</sub> +W <sub>4</sub> )/W <sub>b</sub>                  | = | 1.32  | PASS              |
| <b>[ CASE II ]</b>                                                       |                                                                   |   |       |                   |
| Empty Pipe with backfilling (Installation Stage)                         | (W <sub>p</sub> +W <sub>5</sub> )/W <sub>b</sub>                  | = | 2.00  | PASS              |
| Empty Pipe with backfilling (After Long Period of Operation)             | (W <sub>pl</sub> +W <sub>5</sub> )/W <sub>b</sub>                 | = | 1.91  | PASS              |
| <b>[ CASE III ]</b>                                                      |                                                                   |   |       |                   |
| Pipe with product and backfilling                                        | (W <sub>pl</sub> +W <sub>5</sub> +W <sub>4</sub> )/W <sub>b</sub> | = | 1.95  | PASS              |
| <b>UPWARD FORCE</b>                                                      |                                                                   |   |       |                   |
| Empty Pipe                                                               | W <sub>b</sub> -W <sub>p</sub>                                    | = | -1.34 | KN/m              |
| Pipe with Product                                                        | W <sub>b</sub> -(W <sub>4</sub> +W <sub>p</sub> )                 | = | -1.54 | KN/m              |
| <b>THE PIPELINE DESIGN IS ADEQUATE.</b>                                  |                                                                   |   |       |                   |

## 10. Attachment 2: Pipe Buoyancy Calculation For 36", API- 5L X70, 23.83 mmThick

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| 36", API- 5L X70, 23.83 mm THK PIPELINES (WITHOUT CONCRETE COATING)      |                                                                   |   |        |                   |
|--------------------------------------------------------------------------|-------------------------------------------------------------------|---|--------|-------------------|
| PIPE BUOYANCY CALCULATION                                                |                                                                   |   |        |                   |
| <b>INPUT DATA</b>                                                        |                                                                   |   |        |                   |
| Outside Diameter                                                         | D                                                                 | = | 914    | mm                |
| Pipe Wall Thickness (Installation Stage)                                 | t                                                                 | = | 23.83  | mm                |
| Pipe Wall Thickness (After Long Period of Operation)                     | t <sub>2</sub>                                                    | = | 17.83  | mm                |
| Corrosion Coating Thickness                                              | t <sub>c</sub>                                                    | = | 3      | mm                |
| Concrete Coating Thickness                                               | t <sub>cc</sub>                                                   | = | 0      | mm                |
| Corrosion Allowance (CA)                                                 |                                                                   | = | 6      | mm                |
| Total Diameter                                                           | D <sub>totp</sub>                                                 | = | 920    | mm                |
| Density of Steel                                                         | ρ <sub>s</sub>                                                    | = | 7850   | kg/m <sup>3</sup> |
| Corrosion Coating Density                                                | ρ <sub>c</sub>                                                    | = | 942    | kg/m <sup>3</sup> |
| Concrete Coating Density                                                 | ρ <sub>cc</sub>                                                   | = | 0      | kg/m <sup>3</sup> |
| Concrete Coating Cut-back                                                | L                                                                 | = | 0.33   | m                 |
| Product Density                                                          | ρ <sub>pr</sub>                                                   | = | 80     | kg/m <sup>3</sup> |
| Density of water                                                         | ρ <sub>w</sub>                                                    | = | 1150   | kg/m <sup>3</sup> |
| Factor of safety                                                         | F <sub>s</sub>                                                    | = | 1.25   |                   |
| Burial depth                                                             | d <sub>b</sub>                                                    | = | 1.5    | m                 |
| <b>SOIL DATA</b>                                                         |                                                                   |   |        |                   |
| Water Specific Density                                                   |                                                                   | = | 11.28  | kN/m <sup>3</sup> |
| Dry Soil Density                                                         | ρ <sub>d</sub>                                                    | = | 7.7    | kN/m <sup>3</sup> |
| Saturated Soil Density                                                   | ρ <sub>sat</sub>                                                  | = | 14.40  | kN/m <sup>3</sup> |
| Submerged Density                                                        | ρ <sub>sub</sub>                                                  | = | 3.12   | kN/m <sup>3</sup> |
| <b>OUTPUT</b>                                                            |                                                                   |   |        |                   |
| Bare Pipe Weight (W <sub>1</sub> ) (Installation Stage)                  |                                                                   | = | 5.132  | kN/m              |
| Bare Pipe Weight (W <sub>1L</sub> ) (After Long Period of Operation)     |                                                                   | = | 3.866  | kN/m              |
| Corrosion Coat Weight (W <sub>2</sub> )                                  |                                                                   | = | 0.080  | kN/m              |
| Concrete Coat Weight (W <sub>3</sub> )                                   |                                                                   | = | 0.000  | kN/m              |
| Weight of Product (W <sub>4</sub> )                                      |                                                                   | = | 0.463  | kN/m              |
| Backfill Weight (W <sub>5</sub> )                                        |                                                                   | = | 4.304  | kN/m              |
| Weight of Displaced Water (W <sub>6</sub> )                              |                                                                   | = | 7.500  | kN/m              |
| Buoyancy of pipe (W <sub>b</sub> )                                       |                                                                   | = | 7.500  | kN/m              |
| Total weight of pipe (W <sub>p</sub> ) (Installation Stage)              | W <sub>1</sub> +W <sub>2</sub> +W <sub>3</sub>                    | = | 5.212  | kN/m              |
| Total weight of pipe (W <sub>pl</sub> ) (After Long Period of Operation) | W <sub>1L</sub> +W <sub>2</sub> +W <sub>3</sub>                   | = | 3.946  | kN/m              |
| Submerged weight of pipe (W <sub>sb</sub> )                              | W <sub>p</sub> - W <sub>b</sub>                                   | = | -2.288 | kN/m              |
| <b>FACTOR OF SAFETY</b>                                                  |                                                                   |   |        |                   |
| <b>[ CASE I ]</b>                                                        |                                                                   |   |        |                   |
| Empty Pipe without backfilling (Installation Stage)                      | W <sub>p</sub> /W <sub>b</sub>                                    | = | 0.69   | 'FAIL'            |
| Pipe with product without backfilling                                    | (W <sub>p</sub> +W <sub>4</sub> )/W <sub>b</sub>                  | = | 0.76   | 'FAIL'            |
| <b>[ CASE II ]</b>                                                       |                                                                   |   |        |                   |
| Empty Pipe with backfilling (Installation Stage)                         | (W <sub>p</sub> +W <sub>5</sub> )/W <sub>b</sub>                  | = | 1.27   | PASS              |
| Empty Pipe with backfilling (After Long Period of Operation)             | (W <sub>pl</sub> +W <sub>5</sub> )/W <sub>b</sub>                 | = | 1.10   | 'FAIL'            |
| <b>[ CASE III ]</b>                                                      |                                                                   |   |        |                   |
| Pipe with product and backfilling                                        | (W <sub>pl</sub> +W <sub>5</sub> +W <sub>4</sub> )/W <sub>b</sub> | = | 1.16   | 'FAIL'            |
| <b>UPWARD FORCE</b>                                                      |                                                                   |   |        |                   |
| Empty Pipe                                                               | W <sub>b</sub> -W <sub>p</sub>                                    | = | 2.29   | kN/m              |
| Pipe with Product                                                        | W <sub>b</sub> -(W <sub>4</sub> +W <sub>p</sub> )                 | = | 1.83   | kN/m              |
| <b>THE PIPELINE DESIGN IS INADEQUATE, CONCRETE COATING IS REQUIRED</b>   |                                                                   |   |        |                   |

| 36", API- 5L X70, 23.83mm THK PIPELINES (WITH CONCRETE COATING)          |                                                                   |   |        |                   |
|--------------------------------------------------------------------------|-------------------------------------------------------------------|---|--------|-------------------|
| PIPE BUOYANCY CALCULATION                                                |                                                                   |   |        |                   |
| <b>INPUT DATA</b>                                                        |                                                                   |   |        |                   |
| Outside Diameter                                                         | D                                                                 | = | 914    | mm                |
| Pipe Wall Thickness (Installation Stage)                                 | t                                                                 | = | 23.83  | mm                |
| Pipe Wall Thickness (After Long Period of Operation)                     | t <sub>2</sub>                                                    | = | 17.83  | mm                |
| Corrosion Coating Thickness                                              | t <sub>c</sub>                                                    | = | 3      | mm                |
| Concrete Coating Thickness                                               | t <sub>cc</sub>                                                   | = | 85     | mm                |
| Corrosion Allowance (CA)                                                 |                                                                   | = | 6      | mm                |
| Total Diameter                                                           | D <sub>totp</sub>                                                 | = | 1090.0 | mm                |
| Density of Steel                                                         | ρ <sub>s</sub>                                                    | = | 7850   | kg/m <sup>3</sup> |
| Corrosion Coating Density                                                | ρ <sub>c</sub>                                                    | = | 942    | kg/m <sup>3</sup> |
| Concrete Coating Density                                                 | ρ <sub>cc</sub>                                                   | = | 3040   | kg/m <sup>3</sup> |
| Concrete Coating Cut-back                                                | L                                                                 | = | 0.33   | m                 |
| Product Density                                                          | ρ <sub>pr</sub>                                                   | = | 80     | kg/m <sup>3</sup> |
| Density of water                                                         | ρ <sub>w</sub>                                                    | = | 1150   | kg/m <sup>3</sup> |
| Factor of safety                                                         | F <sub>s</sub>                                                    | = | 1.25   |                   |
| Burial depth                                                             | d <sub>b</sub>                                                    | = | 1.50   | m                 |
| <b>SOIL DATA</b>                                                         |                                                                   |   |        |                   |
| Water Specific Density                                                   |                                                                   | = | 11.28  | kN/m <sup>3</sup> |
| Dry Soil Density                                                         | ρ <sub>d</sub>                                                    | = | 7.7    | kN/m <sup>3</sup> |
| Saturated Soil Density                                                   | ρ <sub>sat</sub>                                                  | = | 14.40  | kN/m <sup>3</sup> |
| Submerged Density                                                        | ρ <sub>sub</sub>                                                  | = | 3.12   | kN/m <sup>3</sup> |
| <b>OUTPUT</b>                                                            |                                                                   |   |        |                   |
| Bare Pipe Weight (W <sub>1</sub> ) (Installation Stage)                  |                                                                   | = | 5.132  | kN/m              |
| Bare Pipe Weight (W <sub>1L</sub> ) (After Long Period of Operation)     |                                                                   | = | 3.866  | kN/m              |
| Corrosion Coat Weight (W <sub>2</sub> )                                  |                                                                   | = | 0.080  | kN/m              |
| Concrete Coat Weight (W <sub>3</sub> )                                   |                                                                   | = | 8.003  | kN/m              |
| Weight of Product (W <sub>4</sub> )                                      |                                                                   | = | 0.463  | kN/m              |
| Backfill Weight (W <sub>5</sub> )                                        |                                                                   | = | 5.099  | kN/m              |
| Weight of Displaced Water (W <sub>6</sub> )                              |                                                                   | = | 10.527 | kN/m              |
| Buoyancy of pipe (W <sub>b</sub> )                                       |                                                                   | = | 10.527 | kN/m              |
| Total weight of pipe (W <sub>p</sub> ) (Installation Stage)              | W <sub>1</sub> +W <sub>2</sub> +W <sub>3</sub>                    | = | 13.215 | kN/m              |
| Total weight of pipe (W <sub>pl</sub> ) (After Long Period of Operation) | W <sub>1L</sub> +W <sub>2</sub> +W <sub>3</sub>                   | = | 11.949 | kN/m              |
| Submerged weight of pipe (W <sub>sb</sub> )                              | W <sub>p</sub> - W <sub>b</sub>                                   | = | 2.688  | kN/m              |
| <b>FACTOR OF SAFETY</b>                                                  |                                                                   |   |        |                   |
| <b>[ CASE I ]</b>                                                        |                                                                   |   |        |                   |
| Empty Pipe without backfilling (Installation Stage)                      | W <sub>p</sub> /W <sub>b</sub>                                    | = | 1.26   | PASS              |
| Pipe with product without backfilling                                    | (W <sub>p</sub> +W <sub>4</sub> )/W <sub>b</sub>                  | = | 1.30   | PASS              |
| <b>[ CASE II ]</b>                                                       |                                                                   |   |        |                   |
| Empty Pipe with backfilling (Installation Stage)                         | (W <sub>p</sub> +W <sub>5</sub> )/W <sub>b</sub>                  | = | 1.74   | PASS              |
| Empty Pipe with backfilling (After Long Period of Operation)             | (W <sub>pl</sub> +W <sub>5</sub> )/W <sub>b</sub>                 | = | 1.62   | PASS              |
| <b>[ CASE III ]</b>                                                      |                                                                   |   |        |                   |
| Pipe with product and backfilling                                        | (W <sub>pl</sub> +W <sub>5</sub> +W <sub>4</sub> )/W <sub>b</sub> | = | 1.66   | PASS              |
| <b>UPWARD FORCE</b>                                                      |                                                                   |   |        |                   |
| Empty Pipe                                                               | W <sub>b</sub> -W <sub>p</sub>                                    | = | -2.69  | kN/m              |
| Pipe with Product                                                        | W <sub>b</sub> -(W <sub>4</sub> +W <sub>p</sub> )                 | = | -3.15  | kN/m              |
| <b>THE PIPELINE DESIGN IS ADEQUATE.</b>                                  |                                                                   |   |        |                   |